

Customer 360 ETL and Data Warehouse Project

1. Introduction

This project implements an ETL (Extract, Transform, Load) process using the Kaggle Customer 360 dataset. The main purpose of the project is to extract data from multiple CSV files, analyze the quality of the data, transform and integrate the datasets, and create a simple data warehouse using a Star Schema.

The project was implemented using Python and Pandas in Google Colab.

2. Dataset

The Customer 360 dataset was obtained from Kaggle.

Dataset: Customer 360

Source: Kaggle

Technology: Python, Pandas, Google Colab

The dataset contains multiple datasets related to customers, orders, and products. These datasets were used to demonstrate multiple-source data extraction and data warehouse development.

3. ETL Process

The ETL process followed in this project consists of the following stages:

Extract → Profile → Preprocess → Transform → Integrate → Data Warehouse → Validate → Export

3.1 Data Extraction

The available CSV files were extracted and loaded separately using Pandas.

The main datasets used include:

- Customers
- Orders
- Products

Each dataset was loaded into a separate Pandas DataFrame.

The original extracted data was kept unchanged during the extraction stage.

3.2 Data Profiling

Data profiling was performed before preprocessing.

The following characteristics were analyzed:

- Number of rows and columns
- Column names
- Data types
- Missing values
- Missing-value percentages
- Duplicate records
- Unique values
- Statistical information

This helped identify data-quality issues before performing transformations.

3.3 Data Preprocessing

Based on the profiling results, the datasets were prepared for further processing.

The preprocessing and preparation included:

- Handling duplicate customer records
- Handling duplicate product records
- Converting date fields into appropriate date formats
- Checking missing and invalid values
- Preparing datasets for integration

Only necessary transformations were applied based on the dataset structure.

4. Data Transformation

Several transformations were performed to prepare the data for the data warehouse.

Important transformations included:

Transformation	Purpose
Date conversion	Convert order dates into a standard date format
Duplicate removal	Remove duplicate customer and product records
Surrogate key generation	Create warehouse-specific dimension keys
Date dimension creation	Support time-based analysis
Foreign-key mapping	Connect fact records with dimension tables
Fact table restructuring	Create an analytical order fact table

The actual order columns available in the dataset were used. The order data contained fields such as `order_id`, `customer_id`, `product_id`, `order_date`, `quantity`, `price`, and `total_amount`.

5. Data Integration

The datasets were integrated using common business keys.

The main relationships used were:

- customer_id
- product_id
- order_date

Customer and product information was mapped to the order data using their respective business keys.

The order date was also mapped to the Date Dimension.

6. Data Warehouse Design

A Star Schema was created for the data warehouse.

The main dimension tables are:

Dimension Tables

dim_customer

Contains customer information and a generated customer_key .

dim_product

Contains product information and a generated product_key .

dim_date

Contains date-related attributes such as:

- date_key
- full_date
- day
- day_name
- week
- month
- month_name
- quarter
- year

Fact Table

fact_orders

Contains order transaction information.

The main fields are:

- order_id
- customer_key
- product_key
- date_key
- quantity
- price
- total_amount

The grain of the fact table is **one row representing one order transaction**.

7. Surrogate Keys

Surrogate keys were generated for the dimension tables.

The following keys were created:

- customer_key
- product_key
- date_key

The original business keys such as `customer_id` and `product_id` were retained where required.

Surrogate keys make it easier to establish relationships between fact and dimension tables in the data warehouse.

8. Data Validation

Several validation checks were performed after creating the warehouse.

The validation process included:

- Checking duplicate keys
- Checking null primary keys
- Checking null foreign keys
- Checking invalid customer keys
- Checking invalid product keys
- Checking invalid date keys
- Checking referential integrity
- Checking negative quantities
- Checking negative prices
- Checking negative total amounts

The `total_amount` field was also compared with:

quantity × price

to identify possible inconsistencies in order totals.

9. Data Warehouse Export

After completing the transformation, integration, and validation stages, the final warehouse tables were exported as CSV files.

The main output tables were:

- `dim_customer.csv`
- `dim_product.csv`
- `dim_date.csv`
- `fact_orders.csv`

These files can be used for further database loading or visualization using tools such as Power BI.

10. Technologies Used

- Python
- Pandas
- Google Colab
- CSV
- SQL
- MySQL/PostgreSQL
- Power BI

11. Conclusion

This project successfully demonstrates a complete ETL workflow using the Customer 360 dataset. Multiple datasets were extracted and profiled before performing preprocessing and transformation. The datasets were then integrated using their business keys and converted into dimension and fact tables.

A Star Schema data warehouse was created using customer, product, and date dimensions together with an order fact table. Finally, data validation checks were performed and the warehouse tables were exported for further analytical use.

This project provides a foundation for future improvements such as database automation, scheduled ETL processing, Power BI dashboards, and automated data-quality monitoring.